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Clinical pathways discovery for long-term and chronic patients: A process mining approach

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HIGHLIGHTS

- Process mining reveals significant variability in clinical pathways for respiratory diseases.
- Increased pathway variability suggests poor adherence to standardized clinical models.
- Episode- and activity-level trajectories identify opportunities for resource optimization.
- Data limitations impact analyses for three diseases, but COPD data provides robust insights.
- Process mining demonstrates potential for improving healthcare delivery and patient outcomes.

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ABSTRACT

The increasing burden of chronic respiratory diseases is placing substantial pressure on healthcare systems around the world. Lung diseases, such as chronic obstructive pulmonary disease, pneumonia, lung cancer, and pulmonary fibrosis, rank among the most prevalent and deadly conditions globally. Given the complexity of managing these chronic conditions, there is an urgent need to optimize care processes to meet the growing service demands efficiently and effectively, especially in public healthcare systems where there is a prevalence of elderly patients. This study aims to understand and improve the quality and performance of treatments provided to patients using a process mining approach. By analyzing clinical data collected from 2018 to 2022, the study identifies critical points in the care pathways where improvements can be made. This approach enables the optimization of resource deployment and service configuration to better meet patient needs. As a case study, this method was applied to a specialized hospital facility dedicated to cardiac and respiratory diseases, where actionable insights were uncovered for enhancing clinical pathways. This study allows us to analyze clinical pathways and detect critical points, providing insights to healthcare managers and decision makers. In addition, it highlights the importance of adequate data collection and suggests that future research efforts should prioritize the acquisition of larger and more diverse data sets to enhance the reliability and validity of activity and episodes analyses.

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1. Introduction

Respiratory diseases such as chronic obstructive pulmonary disease (COPD), pneumonia, lung cancer, and pulmonary fibrosis remain leading contributors to global morbidity and mortality rates [35]. According to the World Health Organization (WHO), COPD was the third leading cause of death globally in 2019, resulting in 3.23 million fatalities [57], while lung cancer ranks as the foremost cause of cancer-related deaths, with the highest death rates in both men and women worldwide [58]. In Canada, respiratory diseases are a significant public health issue, with COPD prevalence hovering around 10% over the past decade [29], and lung cancer projected to be the second most commonly diagnosed cancer in men and women in 2024 [13]. The prevalence of conditions such as COPD and asthma continues to grow, creating substantial challenges for healthcare systems tasked with managing long-term care for affected patients [6].

The growing burden of pulmonary diseases places immense pressure on healthcare systems worldwide. This is due to the complexity of treating chronic illnesses, which require continuous monitoring, specialized interventions, and frequent hospital admissions. Beyond the physical toll on individuals, these diseases lead to significant economic and social challenges, diminishing patients' quality of life and escalating healthcare expenses [46]. As healthcare systems struggle to meet these demands, advancements in Hospital Information Systems (HIS) have become crucial in managing the increasing volume of patient data.

Currently, most of the HIS are able to collect and manage a large amount of data [3]. Within the data collected, we can identify three main categories: clinical, administrative and process data [48]. Clinical data refer to information related to a patient's health status, medical history, diagnoses, treatments, medications, laboratory results, and other details relevant to their healthcare. This data is typically generated and used by healthcare professionals during the provision of care. Clinical data is crucial for making informed medical decisions, assessing patient outcomes, and monitoring health conditions [33]. Administrative data pertain to the operational and financial aspects of healthcare services. It includes information related to billing, scheduling, insurance claims, and general administrative tasks. This type of data is used for managing healthcare facilities, processing payments, and ensuring compliance with regulations. Administrative data help healthcare organizations streamline operations, optimize resource allocation, and maintain accurate records for regulatory purposes [51]. Process data, or operational data, refer to information that tracks the flow of activities, events, or steps within healthcare processes or workflows. It includes data related to the sequence of events, time intervals, workflow efficiency, and performance metrics [48]. While clinical data focus on the medical aspects of patient care, administrative data deal with the operational and financial aspects of healthcare management, and process data track the flow and efficiency of healthcare processes. All three types of data play crucial roles in ensuring effective healthcare delivery and management [56].

In recent years, the international literature has seen an increase in the number of studies focused on the use of process data in healthcare with new techniques, such as process mining [20]. Process mining is a data-driven analytical approach closely associated with business process management (BPM). It focuses on the analysis and visualization of event data recorded during the execution of business processes. By systematically extracting valuable insights from event logs, it enables organizations to uncover the true dynamics of their workflows, pinpoint inefficiencies, detect deviations from intended process designs, and identify opportunities for optimization. This technique bridges the gap between theoretical process models and their real-world execution, contributing significantly to process improvement and performance enhancement [1]. In this study, we present the application of process mining techniques to extract the clinical pathways of chronic and long-term patients with four respiratory diseases treated in a specialized hospital center in Quebec City, Canada.

2. Literature review

In recent years, healthcare management literature has increasingly emphasized the critical role of clinical pathways in enhancing patient outcomes and optimizing the use of resources [38]. Clinical pathways serve as structured frameworks for coordinating patient care, guiding individuals through every phase of their treatment journey—from initial contact to recovery or end-of-life care. With healthcare systems under growing strain, especially following global disruptions such as the COVID-19 pandemic, the need for innovative methods to assess and refine these pathways has never been more urgent.

This review focuses on the integration of process mining techniques into the study of clinical pathways, showcasing its potential to enhance healthcare delivery. We will explore the current research landscape surrounding clinical pathways, particularly in the context of chronic disease management, and discuss how process mining offers new insights into the efficiency and effectiveness of care. By examining recent advancements and challenges, this review aims to deepen our understanding of how process mining can be leveraged to respond to the evolving demands within healthcare systems.

2.1. Process mining in healthcare

Interest in process mining within healthcare has grown in recent years, reflecting the increasing recognition of its potential to improve healthcare delivery [20,22,25,45]. Process mining has emerged as a valuable tool in healthcare settings, with its ability to analyze and enhance various processes within and across medical environments. Process mining tools use event logs from information systems to provide objective, data-based insights into real business processes, offering a clearer picture than assumptions or manual analysis [2]. This granular analysis enhances the transparency and understanding of processes, helping to pinpoint areas for refinement and efficiency improvement [12].

Numerous studies have explored the benefits of applying process mining techniques in this field, emphasizing its capacity to provide insights into patient pathways, resource management, and process inefficiencies [4,8,11,14–16,28,31,32,34,36,39,43,55]. Despite these advantages, the adoption of process mining in healthcare remains somewhat limited, primarily due to the need for detailed and precise process data, which are not always readily available. While hospitals generate significant amounts of data, large-scale, systematic applications of process mining beyond research contexts are still rare, as noted by Munoz-Gama et al. [44,47].

Several studies have addressed the unique challenges faced in applying business process analysis (BPA) and process mining in healthcare environments. Traditional BPA approaches often fall short due to the complex, dynamic, and multidisciplinary nature of healthcare processes, as emphasized by Rebuge and Ferreira [52]. These authors propose a methodology that integrates process mining techniques specifically tailored for healthcare, encompassing data collection, preprocessing, process discovery, and analysis. Their methodology was validated through case studies, demonstrating its capacity to capture the complexity of healthcare processes and support improvements in organizational efficiency. Similarly, Mans et al. [42] identified various challenges such as dispersed data sources and inconsistent timestamp granularity, which complicate the application of process mining in healthcare. They advocate for further research to address these issues, particularly to improve data integration and enhance the utility of process mining techniques.

Process mining has been applied successfully in several specific healthcare contexts, providing valuable insights into patient care and resource management. For example, Llatas et al. [26] explored the use of process mining in conjunction with indoor location systems (ILS) to offer a user-friendly and non-intrusive way to visualize hospital processes. Their approach allowed healthcare professionals to better

understand the workflows and inefficiencies in specific areas, thus facilitating more informed decision-making. Delias et al. [21] proposed a methodology for clustering patient flows and creating concise summaries using process mining, addressing the issue of noise and outliers through a spectral technique. This method was successfully tested in a real healthcare setting, providing clear process maps and offering insights into the complexity of patient flows. Other studies have focused on improving the accuracy and efficiency of process mining in healthcare, through advanced data analysis and preprocessing techniques. Lismont et al. [40] proposed a methodology that helps healthcare organizations analyze patient pathways, workflow, and guideline compliance, overcoming challenges such as patient-specific processes and data format inconsistencies. De Leoni et al. [19] introduced a framework for integrating correlation analysis with process mining, to identify bottlenecks and inefficiencies, emphasizing the importance of correlating different process characteristics for comprehensive analysis. They demonstrated the utility of their approach in a practical case study, underscoring its effectiveness in organizing event logs and extracting process insights.

The issue of data quality has also been a significant concern in healthcare process mining. Suriadi et al. [54] addressed this by proposing models to systematically handle data imperfections, such as incomplete or inaccurate data and timestamp inconsistencies. Their research confirmed the prevalence of such imperfections even in highly mature event logs, highlighting the necessity of rigorous data cleaning before conducting process mining analyses. Along the same lines, Baker et al. [10] developed a methodology for extracting detailed patient care pathways from electronic health records, focusing on cancer patients undergoing chemotherapy. Their iterative process mining approach allowed for the reconstruction of clinical pathways and the collection of robust data, which informed a cost-effective decision-making model for home monitoring of patients.

In summary, while process mining holds great promise for improving healthcare processes, its adoption is still in its early stages. The growing body of research demonstrates its potential across various applications, from visualizing hospital workflows to extracting detailed clinical pathways. However, challenges such as data quality, integration, and process complexity must be addressed to fully realize the benefits of this technique in healthcare.

2.2. Clinical pathways

We focus on the application of process mining to patient clinical pathways. Clinical pathways, also known as care pathways, as defined by Gartner et al. [27] are “a long-term and complex managerial intervention adopting a systemic approach, for a well-defined group of patients who journey through the entire continuum of care, from prevention and screening to recovery or palliative care.” By applying process mining to these pathways, we aim to detect the actual flow of patients through various stages of care, model their clinical pathways, and identify any deviations from the intended care processes.

Understanding and modeling patient clinical pathways are critical for several reasons [38]. It allows healthcare providers to clearly visualize the actual flow of care, providing real-time insights into how patients navigate through the healthcare system. These insights are vital for detecting variations that might indicate deviations from standard procedures or highlight inefficiencies within the care process. Moreover, modeling these pathways ensures that the care patients receive is in line with established best practices and clinical guidelines [9].

Identifying inefficiencies and deviations in clinical pathways enables healthcare organizations to refine their processes, eliminate unnecessary steps, and optimize the use of resources [41]. This optimization can lead to quicker patient throughput and potentially reduce hospital stays. When clinical pathways are enhanced, particularly through the use of process mining techniques, patient care becomes more consistent and effective. This increased consistency can lead to better patient outcomes by reducing variations in care, minimizing errors, and ensuring that treatments are evidence-based [5].

3. Data analysis and methodology

The historical clinical data provided by the hospital enables us to define an empirical pathway, i.e., the clinical pathway resulting from process mining, that represents the real pathway performed for the patients. While these empirical pathways are intended to be juxtaposed against theoretical pathways constructed from protocols and guidelines in the broader scope of the project, this paper focuses exclusively on the extraction and analysis of empirical pathways. The objective is to identify critical junctures and areas for process optimization based on observed patient trajectories. This analysis encompasses two types of process analysis based on different levels of granularity, which are explained in Section 3.2. The pathways are traced for four distinct pulmonary diseases: lung cancer, pneumonia, chronic obstructive pulmonary disease (COPD), and pulmonary fibrosis.

3.1. Ethical process and consent to participate

The organization’s ethics review committee gave a favourable review of the original study on June 23 2022 [Reference no. 2022-3826, 22200]. Several measures were implemented, including obtaining written consent from participants, the method of storing data, commitments to confidentiality, and anonymity of participants. Participation in this study was voluntary.

3.2. Key terms and definitions

In process mining, we distinguish between “case” and “activity” as foundational concepts to evaluate and describe a process [44]. To ensure clarity, we define an “episode” as a significant interaction between a patient and the healthcare system that is related to their chronic disease. For this study, we categorize episodes into three distinct types: hospital stay, emergency department access, and outpatient visits. We then define the two types of process analysis as follows:

- Macro-Level Analysis (or Episode-Level): At this level, the process mining “case” is defined as the long-term treatment of a chronic disease, while the “activities” are the individual episodes that occur during the treatment. We consider the timestamp (date and time) of an episode as the start time of the episode itself. For example, a patient (e.g., Patient X) diagnosed with a specific pathology (e.g., Pathology Y) might experience multiple episodes, such as a hospital stay, followed by an outpatient visit three months later, and subsequently an emergency department access two weeks after that.
- Micro-Level Analysis (or Activity-Level): At this level, the process mining “case” corresponds to a single episode, and the “activities” represent the clinical steps performed during that episode. For instance, for the same patient (e.g., Patient X) hospitalized for Pathology Y, the activities during their hospital stay may include admission, laboratory analysis three hours later, surgery at a subsequent point, and discharge two days afterward.

Table 1 shows a summary of this distinction with examples.

On a macro-level, the analysis involves charting the progression of various episode types across a patient’s clinical journey to comprehend the overall flow and transitions between different healthcare settings. Another goal at this level was to identify the origin of patients who had outpatient clinic visits—that is, the preceding episode type from which the patient transitioned into the outpatient setting—and to calculate the mean duration between that originating episode and the outpatient visit itself. On a micro level, it entails examining the sequences of specific

Table 1
Summary of event log construction at macro and micro levels, with examples.

Level of analysis	Macro - level	Micro - level
PM Case	Patient's long-term treatment	Single episode (e.g., HA)
Activities	HA, ED, OV	Admission, lab test, surgery, discharge
Exemple	Patient X:	Patient X's HA:
	HA → OV → ED	Admission → Lab → Surgery → Discharge

Note: HA = Hospital admission, ED = emergency department episode, OV = Outpatient visit.

activities within each episode type and identifying the delays between different activities.

3.3. Data source and collection

The data provided here were supplied by Institut universitaire de cardiologie et de pneumologie de Québec - Université Laval (IUCPQ-UL), a prominent healthcare organization in Quebec specialized in the health of persons with cardiovascular and respiratory diseases which serves a population base of over 2,2 million, one of the largest respiratory medicine centers in Canada. The data sets cover a diverse range of healthcare information and were collected over a specified period to support the analysis of the optimization of clinical pathways for long-term and chronic patients (see Fig. 2).

Our study utilized four primary databases sourced from IUCPQ-UL, each containing distinct but complementary sets of information:

- **Hospital Admissions Database**: Contains detailed records of patient admissions to hospital facilities, including admission dates, discharge dates, primary diagnoses, and procedures performed during hospital stays;
- **Emergency Department Database**: Includes data on patient visits to emergency departments, capturing information such as presenting complaints, triage assessments, diagnostic tests, and disposition outcomes;
- **Outpatient Visits Database**: Provides records of patient visits to outpatient clinics, comprising appointment dates, medical consultations, appointment types, and main diagnoses;
- **Medical Imaging Database**: Contains reports and findings from diagnostic imaging procedures, including X-rays, CT scans, MRI scans, along with request and realization dates.

The mentioned databases tables share a common identifier, the patient's ID, which allows for the integration and analysis of data across all four sources. The Medical Imaging database presents a unique challenge in its relationship to the other three databases. While the first three databases typically contain records that directly correspond to specific patient episodes, each medical imaging record solely references the patient's ID without a direct link to a particular episode. Consequently, establishing connections between medical imaging records and their corresponding episodes requires a more intricate approach. This challenge requires a careful examination of various data points and patterns within the databases to accurately link each medical imaging record to its associated patient episode. To resolve this issue, we employ a combination of the patient's ID and the date-hour information. By leveraging these two key pieces of data, we can establish temporal correlations between medical imaging records and potential episodes of care recorded in the Hospitalizations, Emergency Department (ED), and outpatient visit databases. This approach allows us to narrow down the possible matches and effectively link each medical imaging record to the corresponding patient episode.

3.4. Data analysis and description

The data collection period spans from January 2018 to December 2022, covering a time frame long enough to capture a comprehensive view of patient journeys within the system.

The raw data provided cover a population of 4714 patients divided among each type of disease, with 970 diagnosed with lung cancer, 1179 with COPD, 285 with fibrosis, and 2280 with pneumonia. Throughout their clinical history, patients have exhibited multiple types of episodes associated with their disease. The data provided allows us to reconstruct the patient's medical history for different types of episodes through the years. Table 2 shows the number of episodes divided by year, disease, and type of episode.

To develop process mining approach, we used Disco, a software by Fluxicon® [24,30], operating under an academic license, to explore both macro and micro clinical pathways within patient clinical histories. Disco allows us to analyze and visualize clinical pathways by automatically generating detailed process maps from event logs. We chose Disco because it is a powerful yet user-friendly tool that effectively represents the information analyzed. Its intuitive interface and robust capabilities make it particularly suitable for our analysis. Disco's ability to handle large datasets and generate insightful visualizations aligned perfectly with our research objectives. We did not consider other algorithms, as Disco's comprehensive features and positive feedback within the academic community reinforced our decision to use it for our study [7,17,18,23,37,49,50,53]. This enables us to capture the actual flow of

Table 2
Distribution of episodes per year, typology and disease.

Disease	Database	2018	2019	2020	2021	2022	Total
COPD	HA	840	925	626	567	374	3332
	ED	1260	1389	981	890	621	5141
	OV	6476	6857	6289	6384	5069	31,075
Cancer	HA	874	1057	804	762	266	3763
	ED	721	807	556	603	332	3019
	OV	13,922	17,650	14,001	14,995	8750	69,318
Fybrosis	HA	139	168	154	151	74	686
	ED	156	234	165	176	92	823
	OV	2275	2826	2442	2041	1099	10,683
Pneumonia	HA	466	614	480	373	176	2109
	ED	641	871	652	525	344	3033
	OV	4379	4922	4717	5233	3760	23,011

Note: HA = Hospital admission, ED = emergency department episode, OV = Outpatient visit.

patient care and scrutinize patient interactions and treatments within individual healthcare episodes.

To ensure transparency and replicability, we provide further details on the configuration of the Disco tool used in our analysis. Disco applies the Fuzzy Miner algorithm, which is designed to simplify complex process models by emphasizing the most significant and correlated elements. In our study, we used the following parameter settings:

- Activity filter: Set to 100%, meaning all recorded activities were included in the process map. This ensured that no activity, regardless of frequency, was excluded from the analysis.
- Path filter: Set to 0%, meaning only the most frequent and significant paths were retained. This resulted in a highly simplified process map that highlights dominant transitions while excluding less common or exceptional flows.

These settings were selected to balance completeness in activity representation with clarity in process flow visualization. By including all activities but limiting the paths to the most frequent ones, we aimed to preserve the full scope of process steps while focusing the analysis on the core behavioral patterns. We acknowledge that different parameter choices can yield substantially different models, and we encourage future studies to explore alternative configurations for comparative insights.

An initial data analysis was developed to navigate the extensive healthcare information and define our focus for information extraction. The analysis began by examining the mean and median duration of hospital stays and emergency department visits across various disease categories. Subsequently, we investigated how different discharge outcomes correlate with the duration of ED episodes. This approach allows us to gain insight into patient care dynamics and obtain a first overview of patient pathways. Table 3 provides an example result for the described analysis showing how the median total duration of an ED episode for a patient with COPD changes as the clinical pathway itself changes. In this study, we opted to use the median instead of the mean to summarize our data, as the distribution of the variables is not normally distributed.

While the differences in median duration between pathway variants may appear small relative to the overall variability, these distinctions were flagged by clinical experts as operationally relevant. The interpretation is therefore grounded more in domain experience and practical implications than in strict statistical significance, particularly given the non-normal distribution of the data.

On a Episode-Level, we were also investigating the mean delays between two types of episode, whether they are identical or distinct. This broader perspective helps in understanding the overall patterns of patient transitions between different healthcare settings, shedding light on the flow of care within the healthcare system. In Table 4 an example of the described investigation is shown: considering patients having a lung cancer diagnosis, we explore the mean delays between two episodes along with the absolute frequency and coverage over the totality of episodes. The absolute frequency refers to the total number of occurrences of a specific process step or path in the entire data set. It counts every single instance of that activity or path, no matter how often it happens within the same episode. The Episode coverage is calculated by dividing the episode frequency of a specific activity or path by the total number of episodes. This metric provides a proportion or percentage representing how many episodes, out of the total, passed through that activity or path.

Table 3
Total median duration for ED episodes, COPD patients.

	Mean duration	Median duration	Standard deviation	Delta to All episodes median
All episodes	14 h 30 m	13 h 12 m	7 h 51 m	/
Episodes ending in hospital admission	14 h 48 m	13 h 18 m	7 h 57 m	+6 m
Episodes ending in normal discharge	14 h 06 m	13 h	7 h 36 m	−12 m
Episodes with at least one specialist visit	15 h 30 m	14 h 48 m	7 h 53 m	+36 m

Table 4
Mean delays between different episodes, lung cancer patients.

Episode 1	Episode 2	Absolute frequency	Episode coverage	Mean delay
ED episode - start	Outpatient visit	302	13.30%	14.6 h
ED episode - end	Outpatient visit	905	39.70%	16.8 d
Hospitalization – admission	Outpatient visit	640	28.10%	2.63 d
Hospitalization – discharge	Outpatient visit	1834	80.50%	26.3 d
Hospitalization – admission	ED episode	2	0.10%	28 m
Hospitalization – discharge	ED episode	209	9.18%	27.3 d
Outpatient visit	ED episode	943	41.40%	30.5 d
Outpatient visit	Hospitalization	1766	77.60%	15.2 d
Outpatient visit	Outpatient visit	2210	97.10%	20.2 d

Note: m = minutes; h = hours; d = days.

The results provide insightful information about the mean delay between recurrent episodes, e.g., the elapsed time between two outpatient visits (last line of table 4), but also allow to detect some unexpected patient clinical pathway behaviors. For instance, nearly one in ten patients accesses the emergency department only 27 days after hospital discharge (line 6 of table 4), and approximately 78% of patients who undergo an outpatient visit access the hospital 15 days after the visit (line 8 of table 4). These results were discussed with medical experts, who identified these behaviors as abnormal and indicative of systemic issues. The problem primarily arises from the shortage of general practitioners responsible for patient follow-up care once the patient leaves the hospital, showing a lack of integrated care between primary and secondary care. This shortage adds strain to emergency departments and hospitals. Implementing better follow-up systems could enhance integration of health care services and patient outcomes, optimizing resource utilization.

As mentioned above, process mining enables the visualization of event log data through process models, and this kind of model is usually represented with workflow maps that reflect the most frequent or representative process behaviors, as interpreted by the underlying mining algorithm. In the case of Disco, this typically involves directly-follows graphs, which prioritize clarity and frequency but may not fully capture concurrent or exceptional behaviors. The resulting maps are shaped by the algorithm's assumptions and representation biases, and should be interpreted accordingly. Fig. 1 represents an example of a possible result of using a process mining tool. In this case, the map represents the activity-level empirical clinical pathway of a patient with pneumonia in emergency department. At this early stage of the analysis, the resulting map appears rather intricate and contains information that may be misleading because it is the result of data that still needs to be cleaned and refined, but it still provides an excellent basis for an initial brief analysis. For the sake of brevity, we included only a sample map in the main text; however, additional maps for other diseases and levels are provided in the annex to ensure a comprehensive presentation of the preliminary analysis.

The process map illustrates, for a given disease, the most frequent paths based on all episodes involving patients with that disease in the emergency department. Thicker lines, or borders, between activities indicate that more cases followed that specific path. Each line is accompanied by the absolute count of episodes that took that route, and this metric is also displayed within each activity. This visual representation makes it easier to identify the most common paths episodes take through the process. Overlaid on the case coverage map is a secondary metric showing the median time spent by cases between activities (edges) or within specific activities (nodes). For nodes, the displayed value

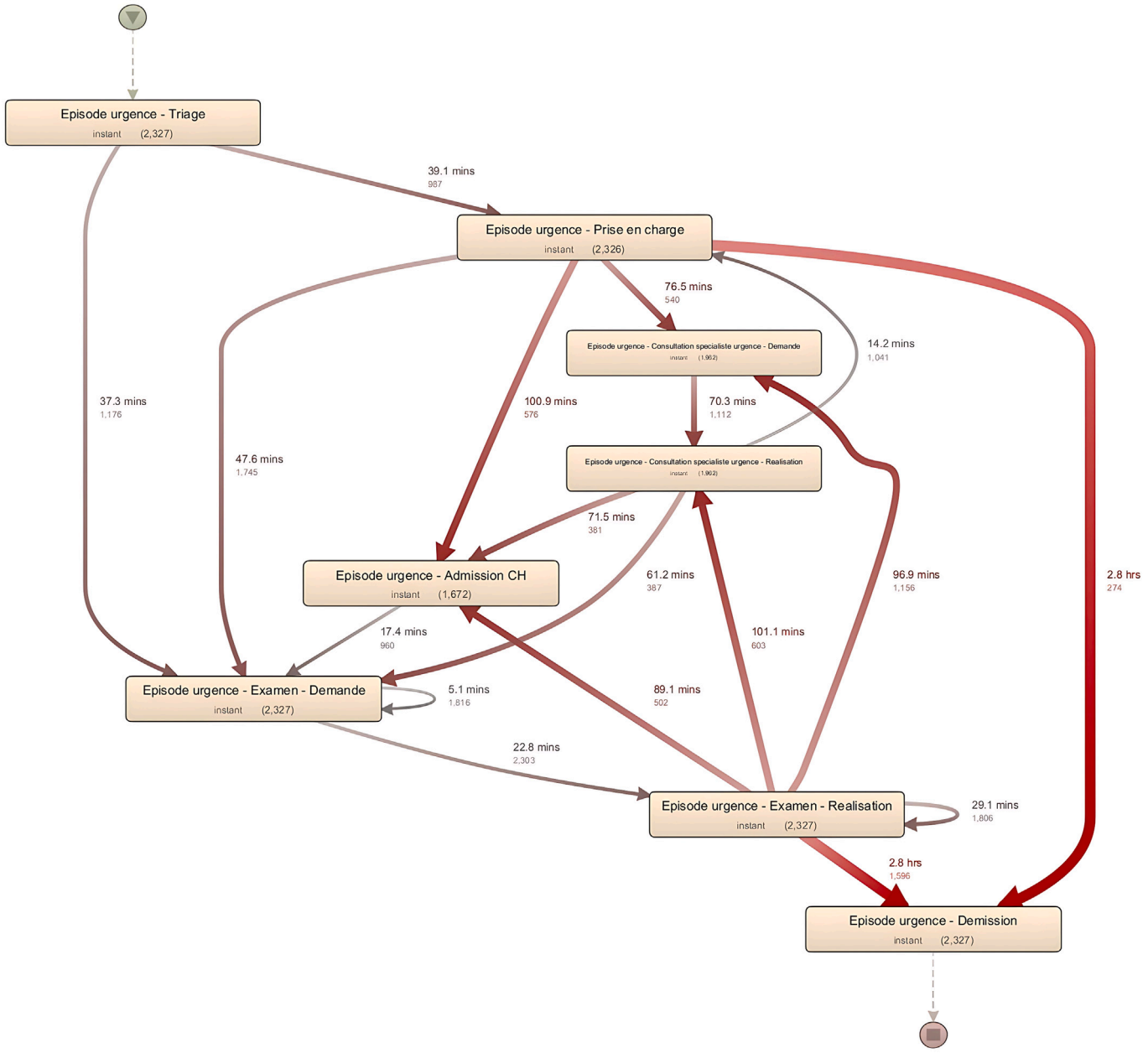


Fig. 1. Activity-level empirical clinical pathway in ED for patients with pneumonia.

represents the median time elapsed between the request and completion of an examination or procedure.

These kinds of maps and tables serve as a foundational framework for our subsequent and more comprehensive analysis described in the following subsection. This deeper analysis encompasses both data cleaning and refinement processes. Through meticulous data cleaning, we ensure the accuracy and consistency of the dataset, while successive refinement involves further exploration and validation of the identified patterns and issues. This iterative approach allows us to iteratively enhance the quality and depth of our insights, facilitating the comparison between the empirical clinical pathway and the theoretical one.

3.5. Information extraction process

At this stage, starting with the preliminary results discussed in the previous section, the project involves multiple rounds of refinement and validation. The process includes several key activities described in Fig. 2. Regular meetings are conducted with domain experts to define and refine the goals of information extraction. These experts, both in the medical and management domains, provide insights that help align project goals with practical needs and expectations.

While we acknowledge the risk of incorporating human bias in such an iterative process, expert involvement is considered essential in scenarios where data quality may be suboptimal. Their domain-specific knowledge enhances the extraction of meaningful and actionable information, complementing the data-driven nature of process mining. To mitigate potential bias, structured consultation frameworks were employed, alongside diverse representation of experts and validation measures based on quantitative metrics. This ensures that the iterative refinement process remains objective while benefiting from the practical relevance provided by expert input. In addition, it is important to clarify that the process mining itself was not iterated directly with expert input. The expert consultations influenced the refinement of the

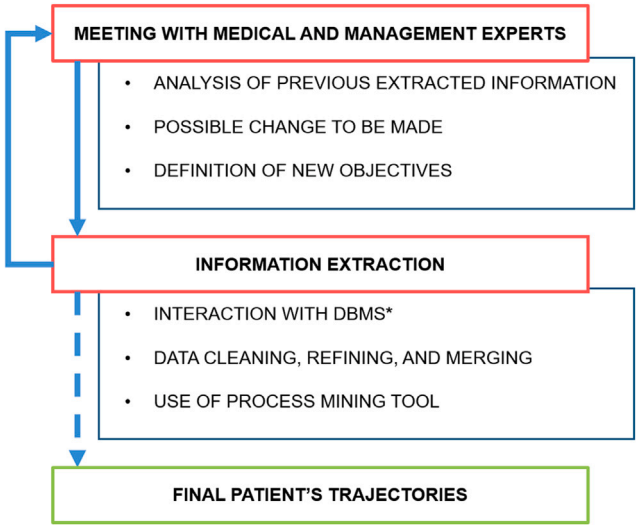


Fig. 2. Information extraction process.

information extraction methods, which served as input for the process mining tool.

The results obtained from each iteration are thoroughly analyzed to understand the effectiveness of current extraction methods and identify any gaps or areas for improvement. Based on analysis and expert feedback, necessary refinements and adjustments are made to information retrieval strategies, which may involve fine-tuning algorithms, adjusting parameters, or incorporating new data sources. This cycle of consultation, analysis, and refinement is repeated several times, with each iteration aimed at improving the accuracy and relevance of the extracted information. The process continues until the extraction and retrieval mechanisms reliably produce the desired patient clinical pathway, achieving the objectives of the project and providing valuable insight for future applications.

During the phase that occurs between two meetings, the information extraction process is divided into two parts: initial extraction and successive refinement, followed by using this information as input for the process mining tool. The first part involves the use of algorithms that interface with the database management system to extract relevant data. These algorithms are responsible for removing outliers, replacing missing data where necessary, filtering data based on dates, activities, IDs, and other criteria, and merging data from different sources or tables. All the algorithms used in this phase are coded in Python (version 3.11). Once the data are prepared, they are then used as input for the process mining tool, which further analyzes and processes the information to derive meaningful insights. This iterative cycle ensures that the extracted information is continually refined and optimized for accuracy and relevance.

4. Results

The described analysis process allows us to explore data from four distinct clinical databases and the common patient ID in these databases facilitated the integration and comprehensive examination of patient care pathways for four pulmonary diseases. The integration of data across the Hospitalizations, ED, and Outpatient visit databases was seamless due to direct references to specific patient episodes. By using the patient's ID and date-hour or timestamps information, we accurately joined imaging records with their corresponding episodes.

At the episodes we aimed to trace the empirical pathway for a patient with a given disease, considering hospital admissions, emergency department episodes, and outpatient visits as key nodes in the overall clinical pathway. These episodes provide essential reference points for understanding the patient's interactions with the healthcare system. To achieve this, we treated individual episodes as PM cases, constructing the event log to include three distinct types of events: "Hospitalisation", captured as the date and time of hospital admission; "Outpatient visit", recorded as the date and time of the outpatient visit; and "ED episode", marked as the date and time of emergency department access. Each event corresponds to one of these nodes in the clinical pathway. Using the process mining tool, we analyzed the sequence, timing, and frequency of transitions between the three event types, generating

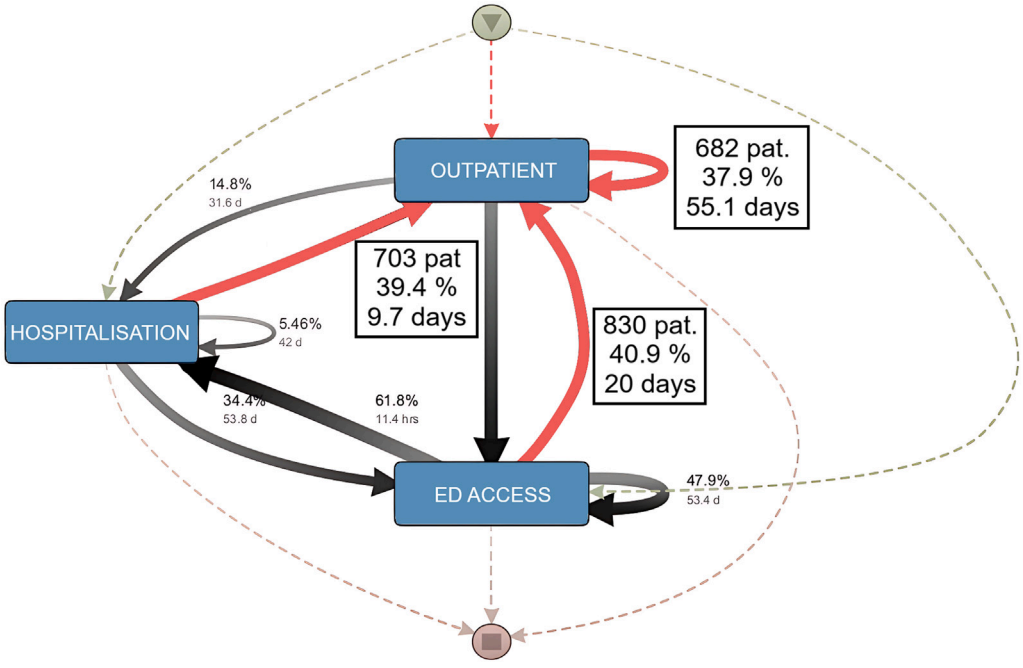


Fig. 3. Output of process mining tool showing the typical episodes level pathway for COPD patients.

a pathway map that reflects the overall clinical pathway. These maps include the median elapsed time between two different activities and the frequencies of the transitions, representing the empirical paths or activities within the process.

An example of these maps is shown in Fig. 3 where it is possible to see the episodes-level pathway of a patient with COPD, there the three types of episodes, the time between them, and the percentage frequency of occurrences of transitions between them are represented. The thickness of the arrows corresponds to the number of occurrences for a given pathway, thicker the arrows, the greater the number of patients passing through that chunk of the pathway. In the red highlighted part of the Fig. 3, we can see that out of the total number of patients with a diagnosis of COPD, 703 (39.4% of the cases) undergo an outpatient visit coming from a hospitalization episode while 830 (40.9%) came from an emergency department episode. We can also see that the median time between a hospitalization and an outpatient visit is 9.7 days, while the median time between an emergency department episode and an outpatient visit is 20 days. We also note that of the total number of 1835 patients who had an outpatient visit, 1600 had more than one visit during their clinical history, while 682 (37.2%) of them went to another outpatient visit without going through another episode, with a median time between the two visits of 55.1 days.

At the activity level, the objective to trace the average pathway of a patient within a hospitalization episode and ED episode is attended by selecting a subset of activities (exams and interventions) performed during episodes that are considered essential and relevant for a patient with a given disease. For each of these activities, we can provide the mean number of occurrences during a single episode and the mean times for the following activities. Specifically, for the 'exam' activity, we considered it as a collection of three distinct sub-activities—'exam request,' 'exam realization,' and 'exam outcome'—which are essential for understanding system behavior. These sub-activities are analyzed at the same level of granularity as the other activities, allowing us to focus on the individual components of each exam occurrence. The following times are calculated:

- T_0: mean time between the beginning of the episode and the first occurrence of the activity request;
- T_1: mean time between the request and the realization of the activity;
- T_2: mean time between realization and outcome of the activity;
- T_3: mean time between the request and the outcome of the activity.
- T_4: mean time between the realization of the activity and the end of the episode;
- T_{inter}: mean time between two realizations of the same activity.

This detailed level of analysis allows for the construction of Fig. 4, which represents a portion of the clinical pathway of a COPD patient within a hospitalization episode. Unlike traditional process maps such as Fig. 1, which provide a general sequence of patient flow and activity connections, Fig. 4 focuses on the temporal dynamics of specific activities. The construction of the pathway relies on detailed information, including the mean duration of a single hospitalization episode, the identification of the first activity occurrence (determined by the lowest T_0), the mean number of occurrences of each activity during an episode, and the mean time intervals between repetitions of the same activity (T_{inter}). Using this information, it was possible to create a clinical pathway that incorporates temporal dynamics and the frequency of activities, offering a detailed representation of a patient's trajectory within a hospitalization episode. Fig. 4 visualizes this pathway, with different colored blocks identifying two distinct activities, such as separate types of exams, while blocks of the same color represent multiple occurrences of the same activity. This granularity enables a nuanced understanding of the efficiency, variability, and temporal progression of care processes.

Tables 5 and 6 provide the times obtained for activities considered relevant during an episode of hospitalization for a patient with COPD. Along with the times described before, for each exam, the tables also provide the absolute frequency ("Abs freq") and the episode coverage (% episode) as defined in Section 3.4, the mean time of occurrence ("Mean occur") as the mean count of occurrences of an exam during an episode, and the percentage of episodes where more than one occurrence is present ("% > 1 occur").

We included only the most representative tables in the main text (Tables 5 and 6). However, additional tables for other diseases are provided in the annex to ensure a comprehensive presentation and understanding of the results.

5. Discussion

Our study utilized a process mining approach to analyze the performance of treatments provided to patients at the Institut Universitaire de Cardiologie et de Pneumologie de Québec - Université Laval in Quebec

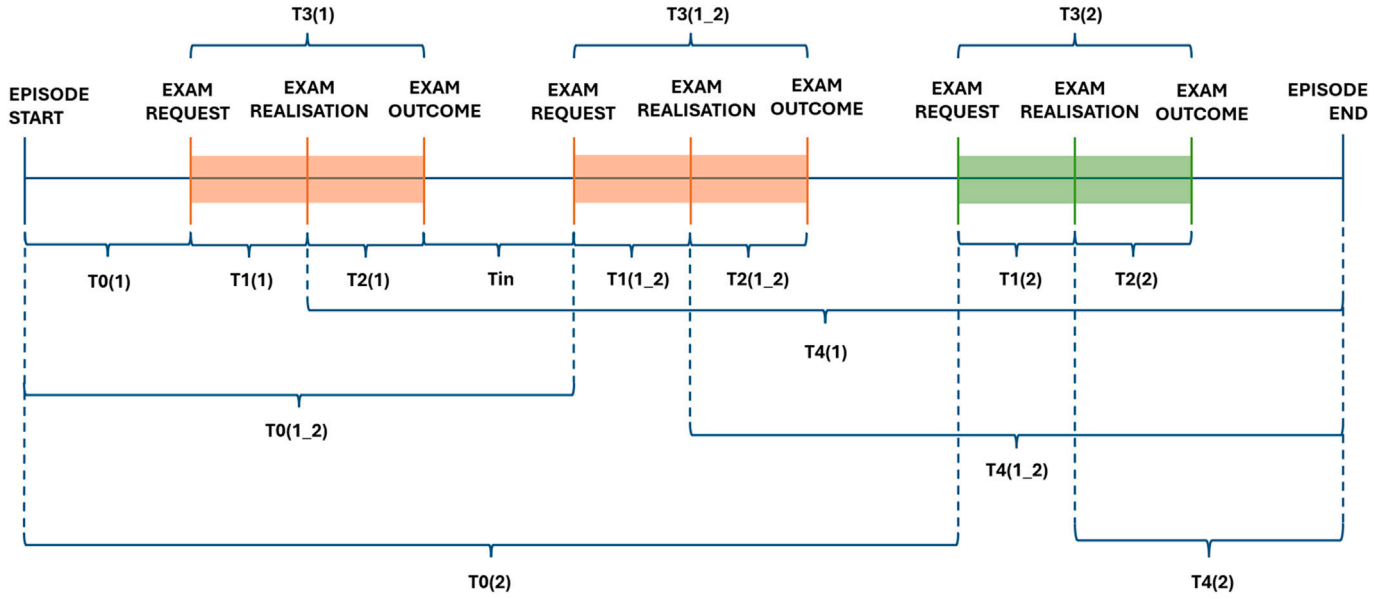


Fig. 4. Portion of a micro-level clinical pathway of a hospitalization episode for a patient with COPD.

Table 5
Resulting times for different exams in hospitalization episode, COPD (1/2).

Exam	T0				T1				T2				T3			
	Mean	Med	Min	Max	Mean	Med	Min	Max	Mean	Med	Min	Max	Mean	Med	Min	Max
[1]	186.07	187.83	87.70	280.92	61.97	66.36	21.25	93.93	34.75	27.89	6.20	77.02	96.72	96.55	52.41	141.36
[2]	320.05	260.16	157.13	542.87	110.35	120.46	20.90	189.70	85.99	45.15	0.38	212.44	196.35	165.61	21.28	402.15
[3]	17.09	3.37	0.72	157.84	2.78	0.72	0.26	21.85	206.22	187.51	7.11	816.56	209.00	188.68	7.57	817.74
[4]	46.41	4.46	0.77	707.51	1.78	0.67	0.18	17.10	194.24	164.36	0.22	816.56	196.02	166.54	0.85	817.74
[5]	35.20	3.87	0.16	1127.46	4.73	0.84	0.03	95.54	226.95	181.20	0.33	2498.10	231.68	184.43	0.56	2499.38
[6]	14.39	2.31	0.33	229.21	1.16	0.62	0.15	10.52	235.03	198.00	17.49	942.38	236.19	198.34	17.74	944.07
[7]	22.34	1.85	0.15	1204.87	1.46	0.54	0.02	94.51	200.22	158.44	0.38	1149.44	201.68	160.13	1.30	1149.75
[8]	61.15	16.84	1.54	782.79	4.42	1.93	0.18	42.54	67.56	28.46	1.03	877.21	71.98	40.30	2.34	882.67
[9]	61.17	15.84	1.54	782.79	4.27	1.93	0.18	42.54	59.63	26.03	1.03	727.11	63.90	35.88	2.34	728.97
[10]	37.94	9.38	0.57	1153.52	3.54	1.05	0.07	67.90	122.09	51.60	0.48	1063.00	125.63	58.23	1.56	1063.95
[11]	43.77	16.77	2.93	331.30	9.56	1.66	0.44	110.95	91.20	24.85	1.00	751.34	100.75	28.08	2.38	752.11
[12]	5.34	5.34	5.34	5.34	0.93	0.93	0.93	0.93	49.51	49.51	49.51	49.51	50.44	50.44	50.44	50.44
[13]	82.60	38.45	1.07	985.23	8.11	1.95	0.05	71.07	86.90	29.84	0.57	580.07	95.01	45.04	1.20	595.15
[14]	161.09	114.10	5.82	325.74	11.50	17.79	0.54	19.66	101.81	71.33	27.07	191.03	113.31	71.87	44.85	210.69
[15]	39.25	39.25	39.25	39.25	24.69	24.69	24.69	24.69	424.72	424.72	424.72	424.72	449.41	449.41	449.41	449.41

[1] Transcutaneous needle biopsy/cytology under ultrasound, [2] Transcutaneous needle biopsy/cytology under CT scan, [3] Left hemithorax, [4] Haemithorax. one or more incidences, [5] Lungs 2 views or less, [6] Lungs 3 or more views, [7] Lungs at the bedside, [8] Lung scan (perfusion with or without flow), [9] Lung ventilation scintigraphy with aerosol or gas-planar and/or tomography, [10] Chest X-ray with contrast, [11] Chest and abdomen X-ray with contrast, [12] Chest and abdomen CT scan without contrast, [13] Chest X-ray without contrast, [14] Chest X-ray without and with contrast, [15] Thorax - scopy.

Table 6
Resulting times for different exams in hospitalization episode, COPD (2/2).

Exam	T4				T inter				Abs freq	% episode	Mean occur	% >1 occur
	Mean	Med	Min	Max	Mean	Med	Min	Max				
1	88.82	77.67	28.72	171.21	–	–	–	–	4	0.07%	1	–
2	215.16	51.69	26.92	566.87	–	–	–	–	3	0.05%	1	–
3	24.24	7.38	0.57	218.25	–	–	–	–	30	0.50%	1	–
4	89.80	10.00	0.16	1240.85	0.05	0.02	0.00	0.16	71	1.19%	1.08	8.45%
5	50.17	13.95	0.00	2570.26	90.98	49.34	0.00	2097.33	3576	59.97%	1.27	12.84%
6	50.33	10.07	0.34	523.33	23.43	23.43	23.43	23.43	40	0.67%	1.03	2.50%
7	63.69	14.48	0.00	1537.52	56.62	28.21	0.00	1029.82	2512	42.13%	1.62	16.56%
8	83.10	13.48	0.57	896.54	–	–	–	–	135	2.26%	1	–
9	79.07	12.37	0.57	896.54	–	–	–	–	126	2.11%	1	–
10	75.21	13.71	0.00	1040.51	235.94	213.31	0.00	575.03	504	8.45%	1.02	1.59%
11	115.97	25.84	0.05	1034.79	–	–	–	–	31	0.52%	1	–
12	3.87	3.87	3.87	3.87	–	–	–	–	1	0.02%	1	–
13	141.25	69.08	0.36	1423.90	203.52	203.21	0.00	549.72	250	4.19%	1.05	4.00%
14	273.20	218.15	0.87	779.51	–	–	–	–	5	0.08%	1	–
15	56.80	56.80	56.80	56.80	–	–	–	–	1	0.02%	1	–

[1] Transcutaneous needle biopsy/cytology under ultrasound, [2] Transcutaneous needle biopsy/cytology under CT scan, [3] Left hemithorax, [4] Haemithorax. one or more incidences, [5] Lungs 2 views or less, [6] Lungs 3 or more views, [7] Lungs at the bedside, [8] Lung scan (perfusion with or without flow), [9] Lung ventilation scintigraphy with aerosol or gas-planar and/or tomography, [10] Chest X-ray with contrast, [11] Chest and abdomen X-ray with contrast, [12] Chest and abdomen CT scan without contrast, [13] Chest X-ray without contrast, [14] Chest X-ray without and with contrast, [15] Thorax - scopy.

City, Canada. The primary focus was on extracting clinical pathways directly from the data, rather than evaluating their conformance to established guidelines. This task will be addressed by other members of the broader project. Our analysis still provides preliminary insights into how these empirical pathways align with the theoretical ones. Specifically, we utilized Disco's 'level-of-detail adjustment' functionality to examine the variability in pathway structures. This involved adjusting the activity level-of-detail, which controls the number of activities displayed in the process map. By focusing on the most frequent activities or including less common ones, we were able to highlight core processes at lower variability levels and uncover exceptional flows at higher levels. In addition, the path level-of-detail was adjusted to manage the number of connections between activities, with fewer paths showing dominant pathways and more paths providing a comprehensive but convoluted view (i.e., "spaghetti process"). Using these tools, we observed that higher variability resulted in maps that were highly convoluted, with no clear structure emerging. This complexity suggests a lack of adherence to a standardized or common pathway. If, even at higher variability levels, pathways remained more distinct and closely aligned to a common model, it would indicate stronger conformance to a unique pathway. The initial findings of poor adherence are likely to be confirmed by future, more detailed analyses from related.

In terms of our study's contribution, the identification of episode-level and activity-level clinical pathways for different diseases revealed opportunities to optimize resource allocation. By leveraging data-driven approaches, our study demonstrated how inefficiencies in hospital processes can be effectively identified and addressed. For instance, the unexpected patient behaviors discussed in Section 3.4, such as the high percentage of patients accessing hospitals shortly after outpatient visits, highlight systemic issues stemming from insufficient integrated care, without a clear continuum of care for the follow-up once the patient leaves the hospital. These findings underscore the importance of integrating process mining methodologies into healthcare optimization strategies. Our findings agree with similar studies in the field [10,40], which underscore the value of process mining for healthcare optimization. While previous research [10] focused on improving cancer episode-level clinical pathways, our study extends these insights by analyzing outpatient, hospitalization, and medical imaging activities on

both episodes and activity scales. This broader perspective reinforces the versatility of process mining in diverse clinical settings.

A critical aspect of our methodology was the information extraction process. To address the challenges posed by low-quality and incomplete datasets, we emphasized the importance of collaborating closely with medical and managerial experts. This interdisciplinary approach enabled us to maximize the value of the available data, ensuring that the extracted insights could provide robust, data-driven support for decision-making. By leveraging domain-specific expertise, researchers can navigate the constraints of imperfect datasets and extract meaningful patterns that directly inform healthcare strategies. This collaboration not only enhances the precision of data interpretation but also empowers decision-makers to implement targeted improvements based on actionable insights.

The implications of these findings are significant for the healthcare system under study, particularly as it faces the challenges posed by an aging population and the rise of chronic diseases. By identifying inefficiencies and areas for improvement, healthcare providers can more effectively allocate resources, reduce costs, and enhance patient care. Not only does this approach support better outcomes for patients, but it also contributes to the long-term sustainability of healthcare services by optimizing the use of available resources.

Nonetheless, this study has several limitations. The quality and quantity of data across the four diseases yielded mixed results. For three of the diseases (Lung Cancer, Pneumonia, and Pulmonary Fibrosis), significant data limitations hindered the reliability of activity analyses, especially due to the lack of detailed exam information and time data. In contrast, for COPD, the data set was robust enough to offer meaningful insights, allowing for confident macro and micro-level analysis. The episodes and activity patterns observed for COPD were consistent and aligned well with the known progression of the disease.

However, the limited data for the other diseases affected our ability to fully recreate the activity clinical pathway. The scarcity of data made it challenging to accurately capture and analyze disease progression, particularly at an activity level. Even the episodes clinical pathways were potentially skewed by the small sample sizes, limiting the strength of our statistical analyses. Therefore, the results for these three diseases should be interpreted with caution, acknowledging the discrepancies between the observed pathways and actual clinical pathways. Part of the variability in the results is also expected due to intrinsic differences between patients, which naturally affect clinical outcomes. Future research with larger and more diverse data sets is essential to validate these findings and improve the reliability of both macro and micro-level analyses.

The limitations of the results are given by the lack of data and its poor quality, highlighting two important suggestions for future research. First, healthcare managers should be encouraged to improve the quality and comprehensiveness of data collection. This would ensure that the resulting data could serve as a robust foundation for more reliable analyses and enable stronger decision-making support. Second, when enhancing data quality proves challenging, it is sometimes necessary to adjust the level of detail in the analysis. This compromise allows researchers to achieve the most accurate insights possible, even from imperfect datasets, by balancing precision with the available data's constraints.

Finally, while our dataset from 2018 to 2022 was comprehensive, it may not fully reflect recent changes in hospital processes or patient demographics. Additionally, since this period was affected by the COVID-19 pandemic, this event could have influenced the results, particularly as healthcare systems worldwide experienced significant disruptions. This study presents a case study of process mining application at a single hospital dataset source for specific diseases; therefore, the findings are inherently context-specific. Broader research across multiple hospitals and diverse datasets would be required to evaluate the framework's applicability in other settings. Broader research that includes additional hospitals and more recent data would help to verify and expand upon our conclusions.

Future studies should also explore the impact of the COVID-19 pandemic on patient clinical pathways, particularly given the widespread disruptions to healthcare systems during this period. Expanding the scope of analysis to encompass other regions, and examining how new healthcare policies and technologies affect patient care, would offer valuable insights into the evolving role of process mining in healthcare.

6. Conclusion

In summary, this study demonstrates that process mining can effectively identify inefficiencies and critical points within clinical pathways at the IUCPQ-UL. This method provides valuable insights into how hospital resources can be better deployed to meet patient needs.

This research contributes to the field by showcasing the applicability of process mining in healthcare settings, particularly for optimizing the care pathways of patients with chronic pulmonary and respiratory diseases. In addition, it adds to the existing literature by providing a comprehensive analysis of multiple clinical activities within a hospital setting.

These findings suggest that hospitals should consider implementing process mining techniques to analyze and improve their clinical pathways. In doing so, they can improve the efficiency and effectiveness of patient care, which ultimately leads to better health outcomes and reduced organizational costs.

This study highlights the importance of leveraging data-driven approaches to optimize healthcare delivery. As the population continues to age and the prevalence of chronic diseases increases, these methods will become increasingly critical in ensuring that healthcare systems remain efficient, effective, and capable of meeting patient needs. Further research and continuous improvement efforts will be essential in realizing the full potential of process mining in healthcare.

While the analysis of COPD provided valuable insights due to the sufficient data quality and quantity, the results for the other considered diseases were significantly compromised by data limitations. These findings highlight the critical importance of adequate data collection and suggest that future research efforts should prioritize the acquisition of larger and more diverse data sets to enhance the reliability and validity of activity and episodes analyses. Moreover, it is imperative to ensure that data collected is of high quality to avoid these types of issues and to provide a more accurate representation of disease clinical pathways. High-quality data collection practices are essential to support robust and reliable research outcomes.

CRediT authorship contribution statement

Luca Murazzano: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Paolo Landa:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Jean–Baptiste Gartner:** Writing – review & editing, Project administration, Investigation, Formal analysis. **Mohamed Hakim Raki:** Writing – review & editing, Investigation, Data curation. **André Côté:** Writing – review & editing, Resources, Project administration, Conceptualization.

Ethics statement

(1) This material is the authors' own original work, which has not been previously published elsewhere. (2) The paper is not currently being considered for publication elsewhere. (3) The paper reflects the authors' own research and analysis in a truthful and complete manner. (4) The paper properly credits the meaningful contributions of co-authors and co-researchers. (5) The results are appropriately placed in the context of prior and existing research. (6) All sources used are properly disclosed (correct citation). Literally copying of text must be indicated as such by using quotation marks and giving proper reference. (7) All authors have been personally and actively involved in substantial

work leading to the paper, and will take public responsibility for its content. (8) Ethical approval for this study/case/case series was obtained from the Institut Universitaire de Cardiologie et Pneumologie de Québec (APPROVAL NUMBER/ID 2022-3826, 22200 – on date June 14th, 2022 – Project name PNEUMO-IUCPQ)* .

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Microsoft Copilot in order to improve the readability and language of the manuscript. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data for this article can be found online at doi:10.1016/j.compbiomed.2026.111539.

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